Unit: Unit 8: Quadratics
Topic: Vertex Form and Intercept Form

Name: ________________
Date: ___________

Advanced (Habanero)

Q1. What is the vertex form of a quadratic function?
(A) $y = a(x-h)^2 + k$
(B) $y = ax^2 + bx + c$
(C) $y = a(x+h)^2 - k$
(D) $x = a(y-k)^2 + h$
(E) $x = a(y+k)^2 - h$

Q2. Which of the following is the vertex form of $y = x^2+4x+4$?
(A) $y = (x+2)^2$
(B) $y = (x-2)^2$
(C) $y = x^2 + 4x + 4$
(D) $y = (x+2)^2 - 4$
(E) $y = (x-2)^2 + 4$

Q3. What does the value of h represent in the vertex form $y = a(x-h)^2 + k$?
(A) the y-intercept
(B) the x-intercept
(C) the vertex's x-coordinate
(D) the vertex's y-coordinate
(E) the axis of symmetry

Q4. What is the intercept form of a quadratic function?
(A) $y = a(x-p)(x-q)$
(B) $y = ax^2 + bx + c$
(C) $y = a(x+p)(x+q)$
(D) $y = a(x-p)(x+q)$
(E) $y = a(x+p)(x-q)$

Q5. Which of the following is the intercept form of $y = x^2 - 5x - 14$?
(A) $y = (x-7)(x+2)$
(B) $y = (x+7)(x-2)$
(C) $y = x^2 - 5x - 14$
(D) $y = (x-2)(x+7)$
(E) $y = (x+2)(x-7)$

Q6. What do the values of p and q represent in the intercept form $y = a(x-p)(x-q)$?
(A) the y-intercepts
(B) the x-intercepts
(C) the vertex's coordinates
(D) the axis of symmetry
(E) the y-intercept and the vertex's x-coordinate

Q7. What is the vertex of the parabola $y = 2(x-1)^2 - 3$?
(A) $(1, -3)$
(B) $(-1, 3)$
(C) $(1, 3)$
(D) $(-1, -3)$
(E) $(3, 1)$

Q8. What are the x-intercepts of the parabola $y = (x+2)(x-4)$?
(A) $x = -2, x = 4$
(B) $x = 2, x = -4$
(C) $x = -2, x = -4$
(D) $x = 2, x = 4$
(E) $x = 0, x = 6$

Open Ended Questions (FRQ)

Q9. Write the quadratic function $y = x^2 + 2x - 6$ in vertex form and intercept form.

Q10. Find the vertex and x-intercepts of the parabola $y = x^2 - 4x - 3$.

Chef's Tips

- Check for the correct use of vertex and intercept forms.
- Verify the accuracy of the calculated vertex and intercepts.
- Ensure that the work is neat, organized, and easy to follow.